The Rationals

Chapter 2 built $\mathbb{Z}$, an ordered commutative ring in which $2x = 1$ has no solution. This chapter supplies the missing quotients by the move that built $\mathbb{Z}$: a pair (p, q) with $q > 0$ records "p divided by q" without performing it, two pairs are identified when $ps = rq$, and a rational number is a class of such pairs. Addition and multiplication are defined on representatives and checked to be well defined, and each field law becomes an identity in $\mathbb{Z}$, already proved. *Sets and Functions*, Section 3.1 took $\mathbb{Q} = \{m/n : m \in \mathbb{Z},\ n \in \mathbb{Z}_{>0}\}$ as a definition; here it is a theorem about the constructed set. The order comes from one choice: a class is positive when a representative has a positive numerator. The Archimedean property then follows from the construction, with no supremum anywhere.

The last section asks what $\mathbb{Q}$ still lacks. No rational squares to 2, and the set of rationals with square below 2 is bounded above but has no supremum in $\mathbb{Q}$. A sequence of rationals, $2, 3/2, 17/12, \ldots$, closes in on the gap from above. Chapters 4 and 5 turn sequences of this kind into the real numbers.

What you need. Chapter 2, all of it, and from Chapter 1 only induction, recursion and well-ordering on $\mathbb{N}$; *Sets and Functions*, Chapters 2, 3 and 5, chiefly Section 5.2; a few results of *Logic and Proof*. From Chapter 2: classes of pairs and the lemma on rules defined on classes (Definition 2.1.1, Theorem 2.2.1), the operations of $\mathbb{Z}$ and the copy of $\mathbb{N}$ in it (Definition 2.2.1, Proposition 2.2.3); commutative rings and fields with their algebra (Definition 2.3.1, Theorem 2.3.1, Proposition 2.3.2), from Section 2.3; ordered rings, orders from a positive set, the order of $\mathbb{Z}$, the rules for inequalities and least elements of sets of integers bounded below (Definition 2.4.1, Proposition 2.4.1, Definition 2.4.2, Theorem 2.4.2, Proposition 2.4.3, Proposition 2.4.4), from Section 2.4; division with remainder and parity (Theorem 2.5.2, Corollary 2.5.3), from Section 2.5. From Chapter 1: Corollary 1.4.3, by which induction (Logic and Proof, Axiom 4.2.1) and the recursion theorem (Sets and Functions, Theorem 3.3.1) apply to $\mathbb{N}$, and well-ordering (Theorem 1.3.3), which Section 3.2 contrasts with $\mathbb{Q}_{>0}$. From *Sets and Functions*: equivalence relations, classes and quotient sets (Sets and Functions, Definition 5.2.1, Sets and Functions, Definition 5.2.2, Sets and Functions, Definition 5.2.4, Sets and Functions, Theorem 5.2.1) and the method box for well-definedness (Sets and Functions, Example (Method: checking that a definition on classes is well defined)), from Section 5.2; upper bounds and suprema (Sets and Functions, Definition 5.3.2, Sets and Functions, Definition 5.3.5), from Section 5.3; functions and their equality (Sets and Functions, Definition 2.1.1, Sets and Functions, Proposition 2.1.1), from Chapter 2; and the smallest-counterexample method with its instance on lowest terms (Sets and Functions, Example (Method: the smallest counterexample), Sets and Functions, Lemma 3.2.4), from Chapter 3. Review exercise R2 uses the countability results of *Sets and Functions*, Chapter 4 (Sets and Functions, Theorem 4.2.6, Sets and Functions, Corollary 4.2.7, Sets and Functions, Proposition 4.2.3, Sets and Functions, Proposition 4.2.4, Sets and Functions, Theorem 4.2.8). From *Logic and Proof*: the negation laws (Logic and Proof, Proposition 2.5.5, Logic and Proof, Proposition 6.2.1, Logic and Proof, Proposition 6.2.3, Logic and Proof, Proposition 6.2.2), for Exercise A1 of Section 3.2 and review exercise R3, and the catalogue of what a proof may not do (Logic and Proof, Example (What is not allowed)), for the find-the-error exercises. Section 3.3 compares its proof with the parity proof that $\sqrt{2}$ is irrational (Logic and Proof, Theorem 2.2.1) and does not use it.

Time. About 15 hours on the core track (all of A, one B and one C per section, the exit checks); about 25 hours for everything.

3.-1.1 Roadmap

- **The Rationals as Classes of Fractions.** Pairs (p, q) with $q > 0$ identified by cross-multiplication, $\mathbb{Q}$ as their quotient set, a sum and a product checked on representatives (the second argument free by symmetry), rules that look like functions and are not, the proof that $\mathbb{Q}$ is a field, and the copy of $\mathbb{Z}$ inside it that turns the class $[(p, q)]$ into the quotient p/q.
- **The Rationals as an Ordered Field; the Archimedean Property.** The positive rationals as the classes with a positive numerator, and why negative denominators would break this; $\mathbb{Q}$ an ordered field with the comparison formula $p/q \leq r/s \iff ps \leq rq$; reciprocals reversing order and the midpoint of two rationals; every rational below a natural number, proved from the construction, with its form $1/n < \varepsilon$; and the integer part $\lfloor x \rfloor$.
- **The Gaps in the Rationals.** No rational squares to 2, proved by descent on a least denominator rather than by parity; the set $\{x \in \mathbb{Q} : x^2 < 2\}$, bounded above with no supremum in $\mathbb{Q}$, each candidate refuted by an explicit rational; and the rational sequence $2, 3/2, 17/12, 577/408, \ldots$ that closes in on the gap from above. The review exercises return to Chapter 2 and to *Sets and Functions* and *Logic and Proof*.

3.-1.2 Named moves

The chapter introduces no new method box; the first four moves below were taught earlier and are run here on new data, and the last is new to this chapter.

- **Check on representatives, one argument at a time** (Sets and Functions, Example (Method: checking that a definition on classes is well defined), with Theorem 2.2.1). Change one representative with the other fixed; each change is one identity at the level below. A formula symmetric in its two inputs gets the second change free. (Section 3.1)
- **Reduce to the level below** (Section 2.3). Write a law about classes on representatives; the two sides become the same pair, or two related pairs, by an identity in $\mathbb{Z}$. (Section 3.1)
- **Order from a positive set** (Proposition 2.4.1). Say which elements are positive, check (P1) and (P2), and read off every rule for inequalities. (Section 3.2)
- **Descent on a least denominator** (Sets and Functions, Example (Method: the smallest counterexample)). Take a counterexample with the least denominator and build one with a smaller denominator. (Section 3.3)
- **Refute a supremum candidate by an explicit rational.** Given a rational s claimed to be the supremum, produce from s an element of the set above it or an upper bound below it. (Section 3.3, the scratch work for Theorem 3.3.2)

3.-1.3 Threads

- **Classes of pairs.** Section 2.1 made integers from pairs of natural numbers, Section 3.1 makes rationals from pairs of integers, and Chapter 5 makes real numbers from classes of sequences of rationals, by the same route.
- **The integer part.** Section 3.2 defines $\lfloor x \rfloor$; for $a \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$, $\lfloor a/n \rfloor$ is the quotient of Section 2.5 (Exercise B3 of Section 3.2), and Section 4.2 builds the integer part in every Archimedean ordered field.
- $1/n < \varepsilon$**.** The corollary of Section 3.2 is the form of the Archimedean property that the limit arguments of Chapters 4 and 5 use, with $\varepsilon \in \mathbb{Q}_{>0}$.
- **The sequence** $2, 3/2, 17/12, \ldots$ Section 3.3 computes it; Section 4.6 proves it Cauchy in $\mathbb{Q}$ with no limit in $\mathbb{Q}$; Chapter 5 opens with it, as the first sequence of rationals that should converge and does not in $\mathbb{Q}$.

3.-1.4 Summary

A rational number is a class of pairs $(p, q) \in \mathbb{Z} \times \mathbb{Z}_{>0}$ under $ps = rq$ (Definition 3.1.1), and the sum and product defined on representatives are well defined (Proposition 3.1.2). With them $\mathbb{Q}$ is a field (Theorem 3.1.3) containing a copy of $\mathbb{Z}$, and every rational is m/n with $m \in \mathbb{Z}$ and $n \in \mathbb{Z}_{>0}$ (Proposition 3.1.4). Declaring a class positive when its numerator is positive makes $\mathbb{Q}$ an ordered field whose order extends that of $\mathbb{Z}$ (Definition 3.2.1, Theorem 3.2.1). Every rational lies below a natural number (Theorem 3.2.3), so for each $\varepsilon \in \mathbb{Q}_{>0}$ some $1/n$ is smaller (Corollary 3.2.4), and every rational has an integer part (Proposition 3.2.5). No rational squares to 2 (Theorem 3.3.1), and $\{x \in \mathbb{Q} : x^2 < 2\}$ is bounded above with no supremum in $\mathbb{Q}$ (Theorem 3.3.2).

What is open: $\mathbb{Q}$ is an Archimedean ordered field with gaps. Can the gaps be filled, and by what? Chapter 4 develops limits in an ordered field to say what "closing in on a gap" means, and Chapter 5 fills the gaps.

3.-1.5 Terms

- **denominator, numerator**: q and p in a representative (p, q) of a rational number (Definition 3.1.1).
- **integer part**: $\lfloor x \rfloor$, the one integer m with $m \leq x < m + 1$ (Proposition 3.2.5).
- **order on** $\mathbb{Q}$: $x < y$ when $y - x$ is positive (Definition 3.2.1).
- **positive rational**: a class $[(p, q)]$ with $p > 0$; the set $P_{\mathbb{Q}} = \mathbb{Q}_{>0}$ (Definition 3.2.1).
- **rational number**: a class $[(p, q)]$ of pairs in $\mathbb{Z} \times \mathbb{Z}_{>0}$ under $ps = rq$ (Definition 3.1.1).